

Anushka Kamerkar

<https://www.linkedin.com/in/anushka-kamerkar/>

US Citizen, Active US Security Clearance

Email : anushkakamerkar@gmail.com

Mobile : +1-812-360-1165

EXPERIENCE

• Avnik Defense Solutions, Inc.

Data Scientist

Oct 2023 - Present

- Automated parsing of unstructured documents (pdf, doc, xlsx, images) using Python libraries (pdfplumber, regex, fuzzywuzzy) to create a comprehensive dataset. Integrated it with Power BI dashboards to track the lifecycle of CDRL submissions, related files and improve decision-making processes.
- Developed a data-driven model using a discrete event simulation software to analyze system operation, RAM, and lifecycle data, conducting post-processing with Python (pandas, sqlite3, matplotlib) to calculate system availability, perform sustainment analysis, enabling stakeholders to identify performance bottlenecks and support decision-making.
- Prepared technical reports and presentations summarizing data-driven insights from analytics projects, utilizing Excel, Power BI for visualizations to effectively communicate trends, key insights and recommendations to senior management.
- Created a RAG-based chatbot leveraging Langgraph, LlamaIndex, and Azure Open AI GPT-4o to query aircraft technical repair manuals, reducing manual search time by 90%, streamlining access to repair information and tools.
- Designed and implemented Docker and Kubernetes-based deployment templates for scalable analytics tools, reducing proof-of-concept to minimum viable product timeline from 2 days to 5 minutes, enhancing rapid experimentation and production deployment efficiency.
- Conducted wire breaking experiment by developing a random forest model to forecast coaxial cable failure with 88% accuracy, analyzing angular force and electrical resistance data, enabling proactive maintenance in aerospace and automotive industries.

• Cherish Health

Machine Learning Engineer

May 2022 - Oct 2023

- Trained and deployed an AutoML model on GCP to predict person activity with accuracy of 91%.
- Leveraged Vertex AI on GCP to build custom model prediction pipelines and endpoints to improve throughput and streamline processes.

• Coditation Systems

Software Development Engineer

July 2019 - July 2021

- Developed a Speech Emotion Recognition system to automate feedback to customer calls. Implemented an ensemble of classification algorithms - RNN, LSTM, ANN to predict emotion of the speaker with accuracy of 87%.
- Scrapers/Crawlers: Optimized content analysis by designing an end-to-end pipeline that utilized EC2 based cron jobs to web scrape over 200TB of data to be ingested into a Snowflake datastore.
- Data Warehousing: Optimized cloud costs by migrating more than 1 petabyte of data to Amazon RedShift from Snowflake

• Coditation Systems

Data Science Intern

Aug 2018 - June 2019

- Created a solution to auto-scale web applications using machine learning concepts. Developed an Artificial Neural Network to identify applications running on a system with a F1- score of 0.92. Developed a Multivariate Regression to create a capacity plan with a F1-score of 0.89.

SKILLS

- Languages:** Python (Pandas, Numpy, SciPy, Scikit-learn, TensorFlow, Keras, Librosa, Matplotlib), R, SQL, Shell script
- Technologies:** Tableau, Power BI, R Studio, Flask, Docker, Kubernetes, Google Cloud Platform, Microsoft Azure, Jira, MS Excel, Git, Snowflake, Amazon Redshift

EDUCATION

• Indiana University, Bloomington

Master of Science in Data Science

Aug. 2021 – May. 2023

• Pune Institute of Computer Technology

Bachelor of Engineering in Computer Science

Aug. 2015 – May. 2019

PROJECTS

- Music Genre Classification:** Performed Exploratory Data Analysis(EDA), data pre-processing and feature extraction from audio files. Developed and trained a Neural Network to classify audio files into various genres with an accuracy 84%.